# Practice workbook · Reading a sine model: values, biggest and smallest

Reading off values of a sinusoidal model or graph by substituting and using exact values and the range of sine and cosine.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-01

# 1. Question 1 · part b.ii · 2 marks

2020 NOV · Paper 01 (regional R)

Worked-example retry · Attempt part b.ii only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

**1** Here is a formula

$$P = 3 + 2\sin\left(\frac{3\pi t}{8}\right) \qquad 0 \leqslant t \leqslant 12$$

(a) Find the exact value of $P$ when $t = \frac{10}{3}$

**(2)**

(b) Find

(i) the largest value of $P$

(ii) the smallest value of $P$

**(2)**

(c) Find the least value of $t$ for which $P = 4$

**(3)**

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## 1. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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## 2. Question 1 · part a · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

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**1** Here is a formula

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## 2. Question 1 · part a · 2 marks · Working and answer

Working and answer

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## 3. Question 1 · part b.i · 0 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b.i only.

DO NOT WRITE IN THIS AREA

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**1** Here is a formula

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**(3)**

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## 3. Question 1 · part b.i · 0 marks · Working and answer

Working and answer

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## 4. Question 12 · part a · 1 mark

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

**12**

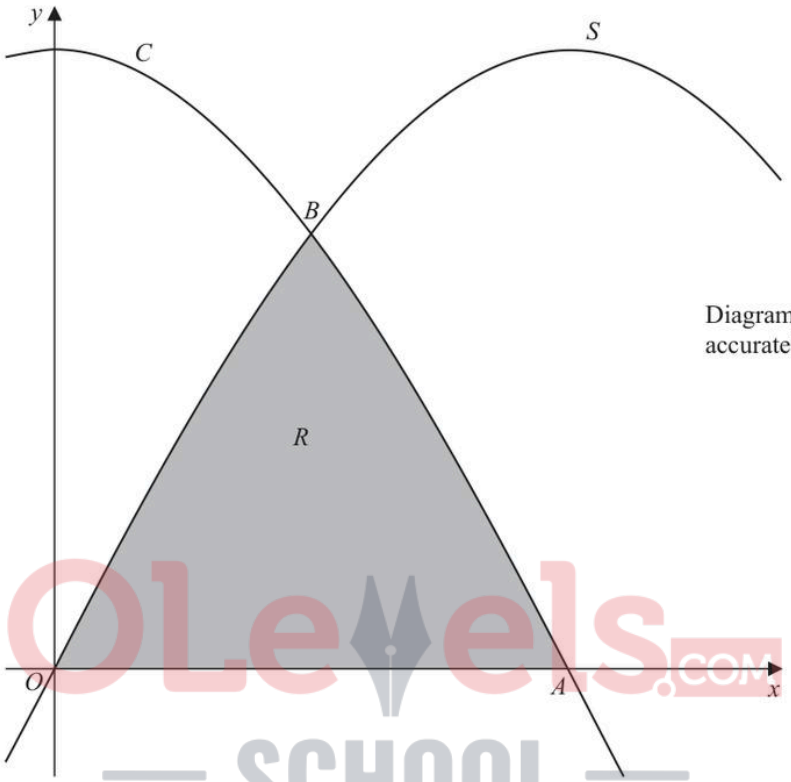



Diagram **NOT** accurately drawn

**Figure 2**

The region $R$, shown shaded in Figure 2, is bounded by the $x$-axis, the curve $S$ with equation $y = 2\sin x$ and the curve $C$ with equation $y = 2\cos x$.
As shown in Figure 2, $C$ crosses the $x$-axis at the point $A$.

(a) Write down the $x$ coordinate of $A$.

**(1)**

As shown in Figure 2, $C$ and $S$ intersect at the point $B$.

(b) Find the $x$ coordinate of $B$.

**(2)**

(c) Using calculus, find the area of the shaded region $R$.
Give your answer in the form $a - \sqrt{b}$ where $a$ and $b$ are integers.

**(4)**

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**38**

P 6 2 2 8 3 A 0 3 8 4 0

## 4. Question 12 · part a · 1 mark · Working and answer

Working and answer

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